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Single-sideband reduced-carrier phase-shifting electro-optic modulator

Abstract

Circuit and method for electro-optic modulation of an RF signal to produce a single-sideband reduced-carrier phase-shifted (SSB-RC-PS) modulated signal includes: receiving an optical carrier signal; receiving the RF signal; modulating the RF signal with a first portion of the optical carrier to produce a single sideband (SSB) modulated signal; amplitude and phase modulating a second portion of the optical carrier signal and combining with the modulated signal to produce a first combined signal, the amplitude modulation and phase modulation of the second portion of the optical carrier signal are dynamically controlled to directly match the amplitude and inversely match the phase of the modulated signal; and amplitude modulating and phase modulating a third portion of the optical carrier signal and combining third portion with the first combined signal to produce the second combined signal, the product of which is a single sideband reduced carrier phase shifted (SSB-RC-PS) modulated signal.

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Background/Summary

FIELD
(1) The present disclosure generally relates to electro-optical circuits and more particularly to a

single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM).

BACKGROUND

(2) Radio-frequency (RF) photonic systems use the strengths of electronics and photonics together to benefit analog RF applications, such as communications and radar applications. Electronics is the dominant technology for analog to digital conversion, digital signal processing, and data storage, while photonics supports terahertz of bandwidth, long-distance low-loss signaling, and simple analog signal processing. RF photonics requires efficient electro-optic modulator (EOM) for modulation of analog electrical signals to the optical domain.

(3) An EOM is a signal-controlled device that modulates laser light based on an electronic analog or digital signal. The modulation may be imposed on the phase, frequency, amplitude, or polarization of the laser light. For conventional EOMs, modulation bandwidths extending into the gigahertz range are possible. Common applications of EOMs include digital data telecommunications and radio frequency (RF) analog signal transmission.

(4) An EOM's performance is quantified by its bandwidth and switching voltage, known as V_pi. One EOM figure of merit is bandwidth/V_pi. Improvements to EOM performance are sought by improving the strength of the electro-optic effect via material science efforts, designing improved electro-optic mode overlap, matching the velocity of electrical and optical waves, and including resonant optical and electronic structures.

SUMMARY

(5) The present disclosure uses optical circuitry to perform lower the effective V_pi of the aggregate circuit. This is done via the present disclosure's single-sideband reduced carrier phase-shifting electro-optic modulator circuitry.

(6) In some embodiments, the disclosure is directed to a circuit and a method for RF interference prediction (detection) and avoidance.

(7) In some embodiments, the disclosure is directed to a single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM). The modulator includes: an optical port for receiving an optical carrier signal; a radio frequency (RF) port for receiving an RF signal; a single-sideband (SSB) modulator for modulating the RF signal with a first portion of the optical carrier, shifting the RF signal to a frequency spectrum of the optical carrier signal and producing a modulated signal; a carrier suppression circuit to amplitude modulate and phase modulate a second portion of the optical carrier to produce an equal amplitude phase shifted optical signal with a phase opposite to phase of the modulated signal. The amplitude and phase modulation of the carrier suppression circuit are dynamically controlled to directly match the amplitude and inversely match the phase of the modulated signal. The modulator further includes: a first combiner to combine the modulated signal from the SSB modulator with the amplitude matched and phase shifted optical signal to produce a first combined optical signal; a carrier injection circuit to amplitude modulate and phase modulate a third portion of the optical carrier to produce an amplitude and phase modulated optical signal, wherein phase and amplitude modulation of the carrier injection circuit is dynamically controlled; and a second combiner to combine the first combined optical signal and the amplitude and phase modulated optical signal to produce a reduced carrier optical signal, as an optical output signal.

(8) In some embodiments, the disclosure is directed to a method for electro-optic modulation of a radio frequency (RF) signal to produce a single-sideband reduced-carrier phase-shifted (SSB-RC-PS) modulated signal. The method includes: receiving an optical carrier signal; receiving the RF signal; modulating the RF signal with a first portion of the optical carrier to produce a single sideband (SSB) modulated signal; amplitude modulating and phase modulating a second portion of the optical carrier signal and combining with the modulated signal to produce a first combined signal, the product of which is a single sideband carrier suppress (SSB-CS) modulated signal, wherein the amplitude modulation and phase modulation of the second portion of the optical carrier signal are dynamically controlled to directly match the amplitude and inversely match the phase of

the modulated signal; and amplitude modulating and phase modulating a third portion of the optical carrier signal and combining third portion with the first combined signal to produce the second combined signal, the product of which is a single sideband reduced carrier phase shifted (SSB-RC-PS) modulated signal.

(9) In some embodiments, the power split ratios for the first portion, second portion, and third portion are dynamically controlled to increase efficiency and reduce optical attenuation. The phase shift on the RF signal is used to accomplish serrodyning. Dynamically controlling the carrier to sideband ratio, modulation depth, effective V_pi, and SOA gain implements an automatic gain control.

(10) In some embodiments, for the third portion of the optical carrier signal, the modulation of the optical carrier is controlled to set the carrier to sideband ratio and modulation depth. This modulation of the optical carrier may further be controlled to set an effective V_pi.

(11) In some embodiments, the phase modulation of the third portion of the optical carrier signal is dynamically controlled to set a static or time-varying RF phase shift, or to set a static or time-varying RF phase shift.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the disclosure. Like reference numerals designate corresponding parts throughout the different views. Embodiments are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which:

(2) FIG. **1** shows a block diagram for a single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM), according to some embodiments of the disclosure.

(3) FIG. **2** depicts a more detailed block diagram and operating principle for a single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM), according to some embodiments of the disclosure.

(4) FIG. **3** is a simplified process flow for electro-optic modulation of a radio frequency (RF) to produce a single-sideband reduced-carrier phase-shifting (SSB-RC-PS) modulated signal, according to some embodiments of the disclosure.

DETAIL DESCRIPTION

(5) In some embodiments, the present disclosure is directed to a single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM). This electro-optic modulator has many benefits for improving EOMs. Carrier reduction increases the signal modulation index--effectively reducing the modulator's switching voltage, known as V_pi. Carrier-reduction, and other integrated circuitry, enable high resolution control of the modulated RF amplitude, and can serve as automatic gain control (AGC). In some embodiments, the architecture supports independent phase shifting of the carrier with respect to the single-sideband. Also, phase shifting can be used to shift the RF sideband's phase, or to accomplish frequency shifting via serrodyning. This enables full control of the RF sideband's phase and amplitude.

(6) Moreover, the circuit architecture makes exclusive use of materials, processes, and components available from existing commercial photonic integrated circuit (PIC), and functions over typical commercial manufacturing tolerances. SSB-RC-PS-EOM performance improves along with intrinsic improvements to EOMs.

(7) FIG. **1** shows a block diagram for a single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM), according to some embodiments of the disclosure. As shown, an optical input port receives an optical (carrier) input signal which is split into three arms (branches); an optical SSB modulator arm, a carrier suppression **113** arm and a carrier injection **119** arm. A

common frequency of the optical signal is in the range of 193.4 Terahertz (THz). The optical SSB modulator is a known (conventional) optical modulator in the art. For example, a SSB modulator can be constructed from an RF hybrid and a dual-parallel Mach Zehnder Modulator. Additionally, methods to implement DC bias control of modulators using detectors **129** and **130** are known in the art. Each of the carrier suppression **113** and carrier injection **119** arms include an amplitude modulator and a phase modulator (**112**/**114** and **118**/**120**, respectively). The amplitude modulators are configured to control the amplitudes of the respective signals in the carrier suppression and carrier injection arms, responsive to respective control signals **126***a* and **128***a*. Similarly, the phase modulators are configured to control the phases of the respective signals in the carrier suppression and carrier injection arms, responsive to respective control signals **126***a* and **128***a*.

(8) In operation, an optical input signal (generated, for example, by a laser) is first (power) split by a splitter**1** into a first light beam **102** and a second light beam **104**. In some embodiments, the power split ratio R**1** of the Splitter**1** is dynamically adjusted in response to control signals **126**, **128**, **129**, and **130** to improve power efficiency and minimize the amount of amplitude attenuation in amplitude modulators **112** and **118**. The first light beam **102** is input to (another) splitter**2** and the second light **104** is input to an amplitude modulator **118** of the carrier injection **113** arm. Splitter**2** (power) divides the first light beam **102** by a ratio of R**2** into a third light beam **106** and a fourth light beam **108**. In some embodiments, the power split ratio R**2** of the Splitter**2** is dynamically adjusted in response to control signals **126**, **128**, **129**, and **130** to improve power efficiency and minimize the amount of amplitude attenuation in amplitude modulators **112** and **118**.

(9) The third light beam **106** is input to the SSB modulator circuit. SSB modulator also takes an RF signal (input to an RF port) as its input, via a dual-parallel Mach-Zehnder modulator (shown as part of the optical SSB modulator). A typical frequency of the RF signal is in the range of DC to 50 GHZ, which is much lower than the frequency of the optical signal. The Mach-Zehnder modulator and RF hybrid perform the optical SSB modulation and shift the RF signal to the frequency spectrum of the optical signal and optically modulate the third light beam **106**, as an output signal **110**. Here, the RF hybrid is a device that equally splits the power of an electrical RF signal into two paths, and shifts one of the paths by 90 degrees, as known in the art. Also, as known in the art, a Mach-Zehnder modulator is a device used to impart a relative phase shift between two signal paths from a single source. The modulator has been used, among other things, to measure phase shifts between the two beams caused by a sample, measure a change in length of one of the paths, or to impart amplitude attenuation on to the optical beam in proportion to an electrical analog or digital data signal.

(10) The fourth light beam **108** is input to an amplitude modulator **112** to control the amplitudes of the fourth light beam **108** in the carrier suppression (CS) **113** arm, responsive to an error signal **126***a* generated by an amplitude/phase control circuit **126**. The amplitude of the fourth light beam **108** is (dynamically) adjusted to match the amplitude of the laser carrier from light beam **110**. The amplitude-modulated fourth light beam **108** is input to a phase modulator **114** to control and shift its phase to an opposite phase of the optical laser carrier in output signal **110**, responsive to the error signal **126***a* generated by the amplitude/phase control circuit **126** to accomplish and monitor carrier suppression. The resulting optical carrier **116** has a phase opposite to that of the phase of the optical carrier in signal **110**. The optical carrier is then cancelled (carrier suppressed) from the modulated signal **110** by combining it with the optical signal **116** with an opposite phase, by a combiner**1** to produce a single-sideband (SSB) carrier suppressed (CS) signal **124**. This signal **124** is also used to generate the error signal **126**A by the amplitude/phase control circuit **126**.

(11) The amplitude and phase of second light **104** are also adjusted by the amplitude modulator **118** and phase modulator **120** in the carrier injection **119** arm, responsive to an error signal **128***a* generated by an amplitude/phase control circuit **128** to achieve control of the injected carrier's amplitude and phase. The resulting carrier injection signal **122** is then combined with signal **124**, by a combiner**2**, to produce a single-sideband reduced carrier (SSB-RC) signal, as an optical

output. In some embodiments, the optical output may be amplified and then converted to an RF signal, for example, by a semiconductor optical amplifier (SOA) and photodiode.
(12) FIG. **2** shows a more detailed block diagram and operating principle for a single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM), according to some embodiments of the disclosure. As shown, an optical (carrier) input signal is generated by a laser **202** and power-split to a first optical signal **230** and a second optical signal **231**, by a splitter **204**. In some embodiments, the input optical signal **202** may be generated by a laser and additional circuitry. An RF signal (RF_in) enters an electrical hybrid **208** where the signal is power split into two arms, with one arm leading the other in phase by 90-degrees. The two RF signals are then input to a dual-drive Mach-Zehnder modulator (DD-MZM) to perform the single-sideband electro-optic modulation. As shown, at the output of the laser and the output of RF-in, the typical frequency of the optical input signal is in the range of 193.4 terahertz, **250**, which is much higher than the typical frequency of the RF signal, which is in the range of DC to 100 GHZ, **251**. The first optical signal is input to another splitter **206** to power-split it into a third optical signal **233** and a fourth optical signal **234**.
(13) The third optical signal **233** is input to another splitter **207** to produce a fifth **235** and sixth **236** optical signal. The sixth optical signal, **236**, is input to the DD-MZM to be combined with the two RF electrical signals resulting from the RF hybrid to produce single-sideband electro-optic modulation as shown in the plot at the output of the DD-MZM, **252**. Note that the suppression of the up-converted lower RF sideband is shown by the dashed grey outline to the left of the optical carrier in **252**. Only the optical carrier and single upper sideband are resultant in signal **237**, shown in **252**. The sixth optical signal **237** is input to another splitter **210** to produce a seventh optical signal **238** and a ninth **239** optical signal.
(14) The fifth optical signal **235** and the seventh optical signal **238** are input to photodiodes PD_B**1** and PD_B**2** respectively, and are used to control the direct current (DC) bias of the DD-MZM, which is known to the prior art. The DC bias inputs to the DD-MZM, which are known to the prior art, are not shown. PD_B**2** is also used in the present disclosure to estimate the optical carrier power of optical signal **239**.
(15) The fourth optical signal **234** from splitter **206** is used to suppress the optical carrier and is input to an amplitude modulator AM_CS to (dynamically) adjust its amplitude, as depicted in plot **253** at the output of the AM_CS. Note that the solid line has a reduce amplitude compared with the dashed grey line, indicating amplitude control. Also note that the amplitude resulting from AM_CS is controlled until the magnitude of the optical carrier shown by the blue arrow in plot **253** is equal to the magnitude of the blue arrow in plot **252**. The amplitude-adjusted fourth optical signal **234** is then input to a phase modulator PM_CS to (dynamically) adjust its phase. The PM_CS is controlled to produce a 180 degree phase shift between optical carrier signals **239** and **240**, as depicted by the upside down blue arrow **254** in the plot at the output of the PM_CS, **254**.
(16) The ninth optical signal **239** output from splitter **210** is combined with the output of the PM_CS **240** by a combiner **212** to generate a first combined signal **241**, as shown in plots **255** and **256**. Note that plot **255** shows the blue carrier signal from plot **252** versus the equal amplitude and opposite phase carrier signal from plot **254**, which results in carrier suppression depicted in plot **256**, where the blue optical carrier has been suppressed, and where only the upper RF sideband shown in red remains.
(17) The second optical signal **231** is input to amplitude modulator AM_CI, to reduce the amplitude of the second optical signal **231** and facilitate control of the carrier injection amplitude, as shown at the output of AM_CI in plot **257**. The output of the AM_CI modulator is input to a phase modulator PM_CI to (dynamically) adjust its phase, or when a repeating phase ramp is input to PM_CI in the case of serrodyning, to (dynamically) adjust its frequency, as depicted by the frequency translated optical carrier shown in plot **258**.
(18) The output of combiner **212** is power-split by splitter **214** to facilitate photodiode monitoring

by PD_CS. PD_CS is used to control the input to AM_CS and PM_CS to facilitate carrier suppression. This way, maximum carrier suppression is achieved by tuning AM_CS and PM_CS until the PD_CS signal is minimized. By including both the amplitude modulator and phase modulator on a single control circuit, any non-ideal phase response in the amplitude modulator, or non-ideal amplitude response in the phase modulator, can be linearized by the control signal and digital signal processing. Additionally, in some embodiments, PD_CI and the attenuation control signal for AM_CS can be used to dynamically control the tunable split ratio of splitter **206** until the attenuation control signal for AM_CS is minimized, and the proportion of power transmitted down path **233** is maximized. Power efficiency is maximized and signal fidelity is maximized when the maximal power is transmitted down path **233** while simultaneously suppressing the carrier using path **234**.

(19) Similarly, the output of PM_CI is power-split by splitter **216** to facilitate photodiode monitoring by PD_CI. PD_CI is used to control the input to AM_CI to facilitate carrier injection. The control signal to AM_CI sets the injected carrier power, which in turn controls the carrier to sideband ratio, which is analogous to modulation depth or the EOM's effective V_pi. The power detected by PD_CI is the injected carrier power. The power detected by PD_CS, when carrier suppression is maximal, is the RF sideband power. The ratio of the carrier to sideband power is then independently controlled by setting AM_CI. PM_CI is used to set the relative phase between the carrier and the sideband. In some embodiments, the sideband phase can be detected based on the output of PD_RF, and used to create a control signal to PM_CI.

(20) A signal from splitter **214** is combined with a signal from splitter **216**, by a combiner **218** to generate a second combined signal, as shown in plot **259** at the output of the combiner **218**. The ratio of the carrier to sideband power is called out in plot **259**, where the difference between the carrier and sideband power is reduced (modulation depth increased), compared with that of plot **252**. Again, this tunable V_pi (tunable modulation depth) is accomplished by controlling AM_CI.

(21) The power estimation of optical signals **239**, **240**, **241**, and **242** can be used in some embodiments to dynamically control the split ratio of tunable splitters **204**, **206**, and **207** to increase optical and electrical RF efficiency, and reduce the need for amplitude attention in amplitude modulators AM_CS and AM_CI.

(22) In some embodiments, the output of the combiner **218** is the amplified by a semiconductor optical amplifier (SOA) **220** with the resulting signal plotted in **260**. Note that, for an equivalent SOA, the present disclosure can increase the sideband power via linear amplification (before compression) beyond what would otherwise be possible due to the present disclosure's increased modulation depth. In some embodiments, a splitter **221** and photodiode PD_SOA are used to monitor the power produced by the SOA.

(23) In some embodiments, the signal existing the SOA from splitter **221** is converted to an electrical RF signal by a photodiode PD_**1** to generate the RF_out signal, shown in plot **261**. In some embodiments, the RF signal frequency can be shifted (shown in plot **261**), the RF phase can be shifted (not shown in plot **261**), and the RF signal amplitude can be dynamically increased (shown in plot **261**) or decreased (not shown in plot **261**) with respect to the input RF signal (dashed grey line in plot **261**, corresponding to original RF signal in plot **251**). The more detailed block diagram of FIG. **2** includes several optional components (compare to FIG. **1**) to further enhance and condition the signals. In FIG. **2**, the DD-MZM represents the first arm (branch), i.e., the optical SSB modulator of FIG. **1**. Similarly, the AM_CS and PM_CS represent the carrier suppression arm (branch) and AM_CI and PM_CI represent the carrier injection arm (branch).

(24) FIG. **3** is a simplified process flow for electro-optic modulation of a radio frequency (RF) signal to produce a single-sideband reduced-carrier phase-shifted (SSB-RC-PS) modulated signal, according to some embodiments of the disclosure. As shown in block **302**, an optical carrier signal is received, for example, from a laser and in block **304** the RF signal is received. In block **306**, a first portion of the optical carrier and the RF_in signal produce a single sideband (SSB) modulated

signal using the RF hybrid and DD-MZM.

(25) In block **308**, a single sideband carrier suppressed (SSB-CS) signal is produced. A second portion of the optical carrier is amplitude modulated and phase modulated. The second signal is combined with the first. The amplitude and phase modulation of the second portion of the optical carrier is controlled until the optical carrier from the first portion is maximally suppressed (e.g., at the output of combiner **212**). Maximal suppression occurs when the carrier amplitude in the first and second paths are perfectly equal, and the carrier phase in the first and second paths are perfectly opposite. The amplitude and phase modulation of the second portion of the optical carrier signal is dynamically controlled to maximally suppress the carrier. An exemplary signal plot is shown by plots **252**, **253**, **254**, **255**, and **256**, in FIG. **2**.

(26) In block **310**, a single sideband reduced carrier (SSB-RC) signal is produced, and the RF sideband is phase shifted. A third portion of the optical carrier is amplitude modulated and phase modulated. The third portion is combined with the product of the combined first and second portions at combiner **218**. Two tasks are accomplished using the third portion of the optical carrier.

(27) In the first task, block **312**, amplitude modulation of the third portion of the optical carrier sets the reduced carrier power. The reduced carrier power sets the sideband modulation depth. The sideband modulation depth is analogous to the modulator's effective V_pi, which is a key figure of merit for modulators. An exemplary signal plot is shown in FIG. **2**.

(28) The second task of phase modulation in block **314**, sets the relative phase difference between the optical carrier and sideband. The relative phase difference between the optical carrier and sideband sets the phase shift of the photodetected electrical RF signal. A static phase shift shifts the phase of the photo-detected RF signal. A phase ramp produces serrodyning and frequency shifts the RF signal. An exemplary signal plot is shown in FIG. **2**. Note the carrier's frequency plotted in **258**, with the resulting frequency of **259** compared to the original carrier frequency (dashed arrow), show serrodyning.

(29) In some embodiment, the optical output signal is amplified and/or converted to an RF output signal. In some embodiment, the monitor signals from PD_B**2**, PD_CS, and PD_CI, along with control signals to AM_CS and AM_CI are utilized to dynamically controlled the power splitting ratio of tunable splitters **204** and **206** to increase efficiency and decrease the attenuation in modulators AM_CS and AM_CI.

(30) It will be recognized by those skilled in the art that various modifications may be made to the illustrated and other embodiments of the disclosure described above, without departing from the broad inventive scope thereof. It will be understood therefore that the disclosure is not limited to the particular embodiments or arrangements disclosed, but is rather intended to cover any changes, adaptations or modifications which are within the scope and spirit of the disclosure as defined by the appended claims and drawings.

Claims

1. A single-sideband reduced-carrier phase-shifting electro-optic modulator (SSB-RC-PS-EOM) comprising: an optical port for receiving an optical carrier signal; a radio frequency (RF) port for receiving an RF signal; a single-sideband (SSB) modulator for modulating the RF signal with a first portion of the optical carrier signal, shifting the RF signal to a frequency spectrum of the optical carrier signal and producing a modulated signal; a carrier suppression circuit to amplitude modulate and phase modulate a second portion of the optical carrier signal to produce an equal amplitude phase shifted optical signal with a phase opposite to phase of the modulated signal, wherein the amplitude and phase modulation of the carrier suppression circuit are dynamically controlled to directly match an amplitude and inversely match a phase of the modulated signal; a first combiner to combine the modulated signal from the SSB modulator with the equal amplitude phase shifted optical signal to produce a first combined optical signal; a carrier injection circuit to

amplitude modulate and phase modulate a third portion of the optical carrier signal to produce an amplitude and phase modulated optical signal, wherein phase and amplitude modulation of the carrier injection circuit is dynamically controlled; and a second combiner to combine the first combined optical signal and the amplitude and phase modulated optical signal to produce a reduced carrier optical signal as an optical output signal.

2. The SSB-RC-PS-EOM of claim 1, further comprising an optical to electrical converter to convert the optical output signal to an RF output signal.

3. The SSB-RC-PS-EOM of claim 1, further comprising an optical amplifier to amplify the optical output signal.

4. The SSB-RC-PS-EOM of claim 1, further comprising tunable optical splitters to dynamically control split ratios.

5. A method for electro-optic modulation of a radio frequency (RF) signal to produce a single-sideband reduced-carrier phase-shifted (SSB-RC-PS) modulated signal, the method comprising: receiving an optical carrier signal; receiving the RF signal; modulating the RF signal with a first portion of the optical carrier signal to produce a single-sideband (SSB) modulated signal; amplitude modulating and phase modulating a second portion of the optical carrier signal and combining with the SSB modulated signal to produce a first combined signal, a product of which is a single-sideband carrier-suppressed (SSB-CS) modulated signal, wherein the amplitude modulation and phase modulation of the second portion of the optical carrier signal are dynamically controlled to directly match an amplitude and inversely match a phase of the SSB modulated signal; and amplitude modulating and phase modulating a third portion of the optical carrier signal and combining with the first combined signal to produce a second combined signal, a product of which is the SSB-RC-PS modulated signal.

6. The method of claim 5, further comprising converting the SSB-RC-PS modulated signal to an RF output signal.

7. The method of claim 5, further comprising amplifying the SSB-RC-PS modulated signal.

8. The method of claim 5, wherein power split ratios for the first portion, second portion, and third portion are dynamically controlled to increase efficiency and reduce optical attenuation.

9. The method of claim 5, wherein a phase shift on the RF signal is used to accomplish serrodyning.

10. The method of claim 5, wherein dynamically controlling a carrier-to-sideband ratio, modulation depth, effective V_pi, and SOA gain implements an automatic gain control.

11. The method of claim 5, wherein for the third portion of the optical carrier signal, the modulation of the optical carrier signal is controlled to set a carrier-to-sideband ratio and modulation depth.

12. The method of claim 11, wherein for the third portion, the modulation of the optical carrier signal is further controlled to set an effective V_pi.

13. The method of claim 5, wherein the phase modulation of the third portion of the optical carrier signal is dynamically controlled to set a static or time-varying RF phase shift.

14. The SSB-RC-PS-EOM of claim 1, wherein the phase modulation of the third portion of the optical carrier signal is dynamically controlled to set a static or time-varying RF phase shift.